

MolPallette — Corpus EDA

naveja_recap

/home/mogan/preprocessed/molpallette/coconut-flavordb_v3/naveja_recap

generated 2026-08-20 09:41

A • Provenance

field	value
sources	['flavordb', 'coconut']
method	naveja_recap
decomposition_params	{'ratio': 0.4, 'include_ring': True, 'max_cores': 4}
core_ratio	0.4
max_cores	4
max_rgroups	8
keep_stereo	True
neutralise	False
dedup	True
n_duplicates_removed	4412
size_filter	{'min_heavy_atoms': 5, 'max_heavy_atoms': 50}
graph_hash_version	3
rdkit_version	2026.03.2
created_at	2026-08-20T09:39:40

B · Composition

quantity	value
molecules	413,458
from coconut	391,964 (94.8%)
from flavordb	21,494 (5.2%)
decompositions	1,344,980
per molecule	mean 3.25, median 4, max 4
R-group slots	2,532,793
per decomposition	mean 1.88
k >= 2	39.19% of decompositions

k>=2 is what the islinked subset space and the core-decoration objective have to work with.

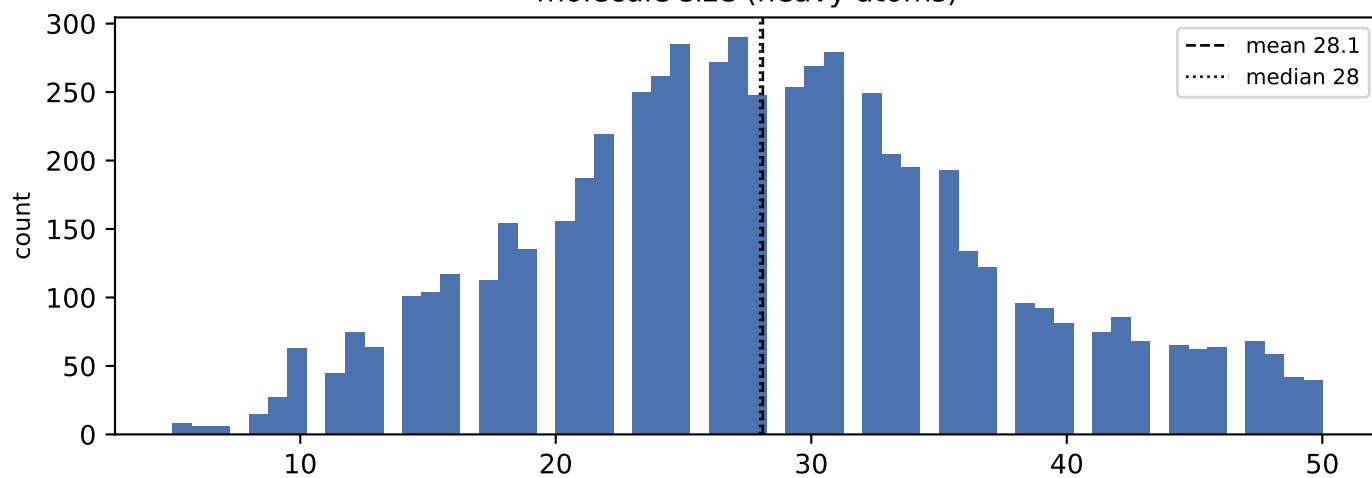
C · Augmentation space

scheme	size
sampling_unit = molecule	413,458 instances/epoch
sampling_unit = decomposition	1,344,980 instances/epoch (3.3x)
full (mol, core, islinked) space	13,479,894 (32.6 per molecule)

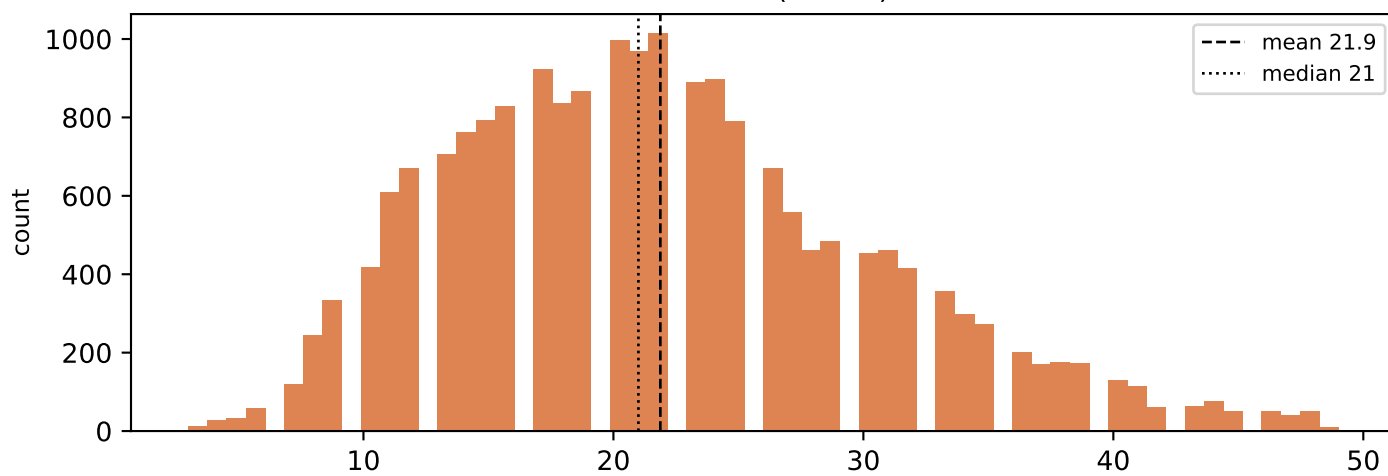
MolPLA's two stages: one molecule yields many cores; one core yields $2^k - 1$ islinked subsets.

D • Size distributions

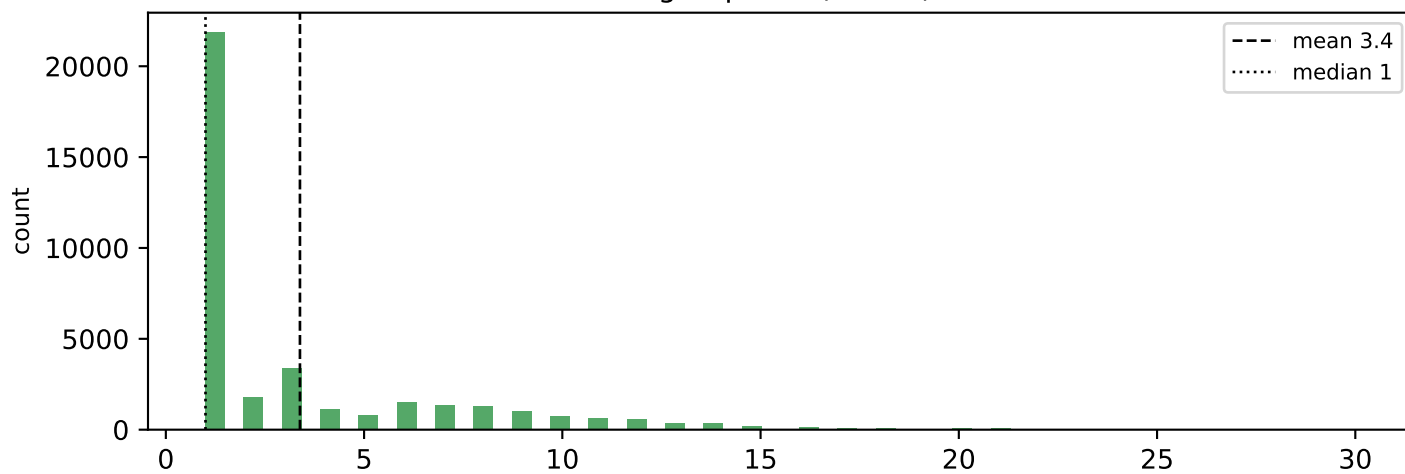
molecule size (heavy atoms)



core size (atoms)

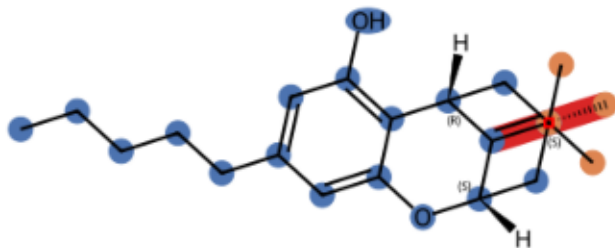


R-group size (atoms)

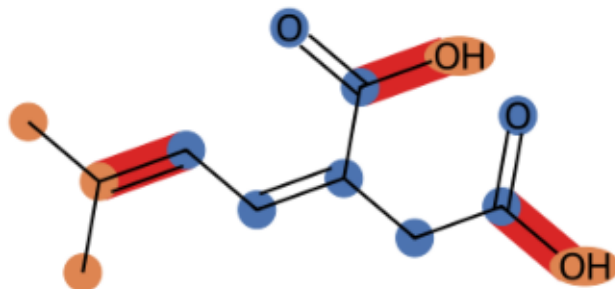


E · Example decompositions (ratio 0.4)

CNP0000074 k=2 core 19 atoms, R-groups [1, 3]



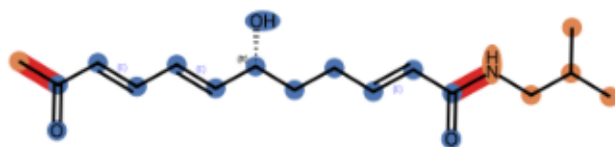
CNP0000161 k=3 core 8 atoms, R-groups [3, 1, 1]



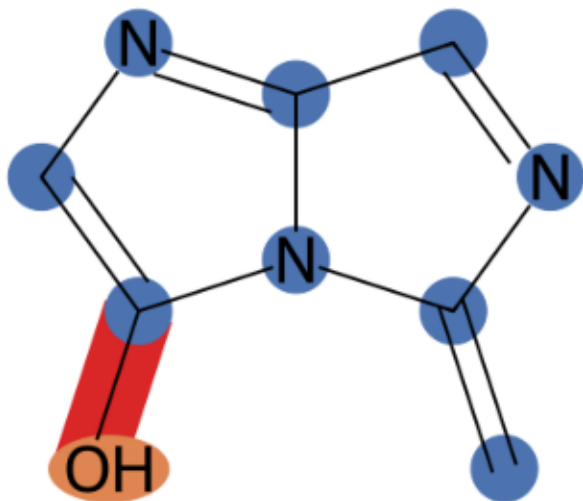
CNP0000233 k=1 core 20 atoms, R-groups [1]



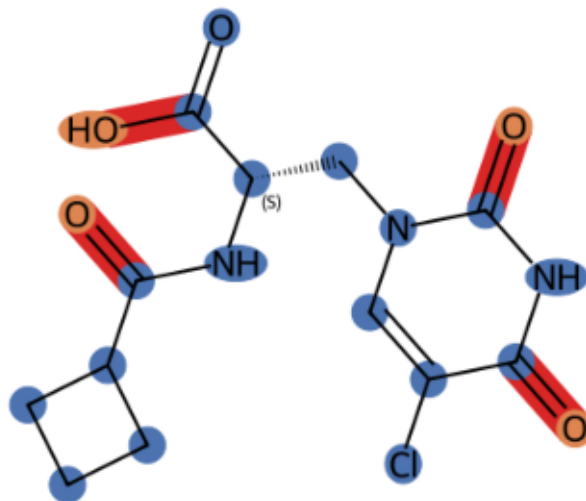
CNP0000307 k=2 core 14 atoms, R-groups [1, 5]



CNP0000386 k=1 core 9 atoms, R-groups [1]

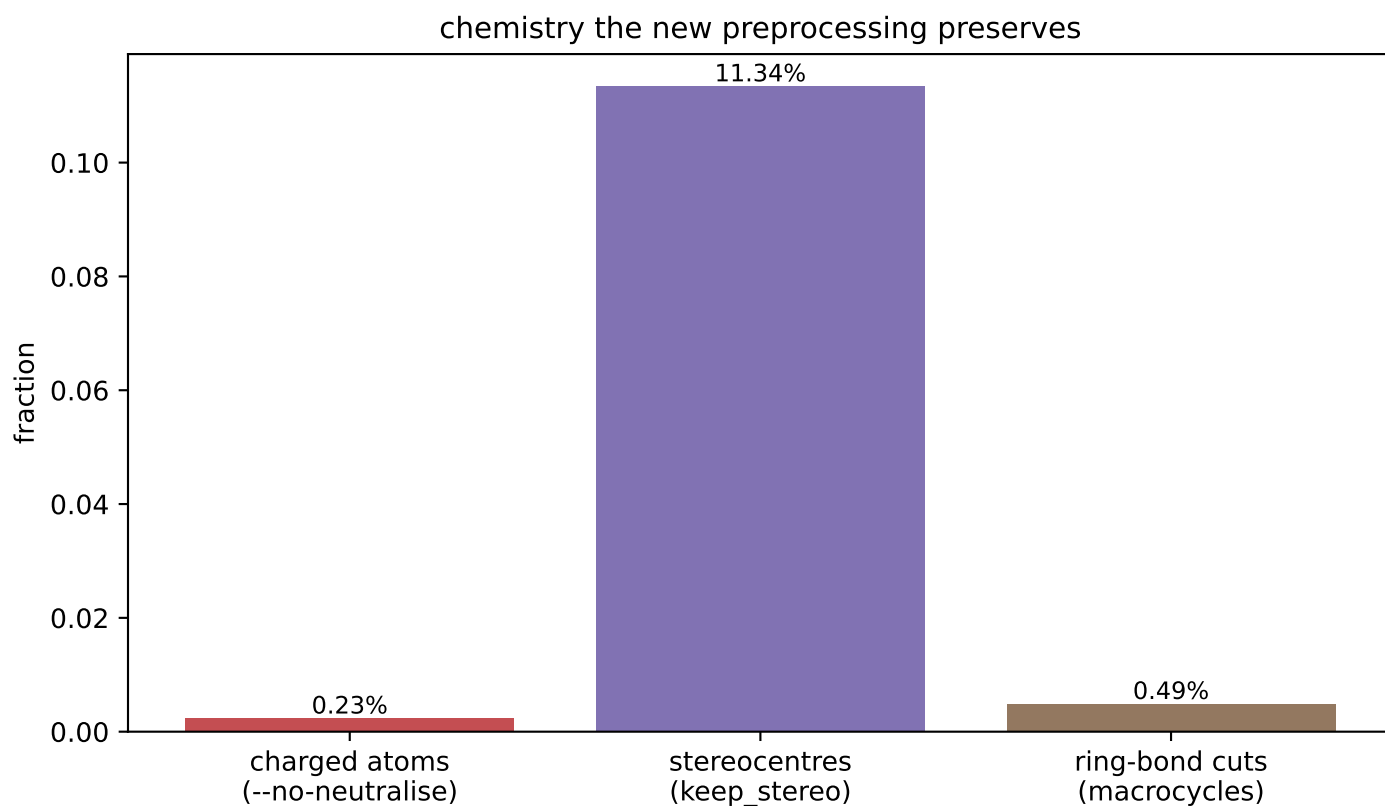
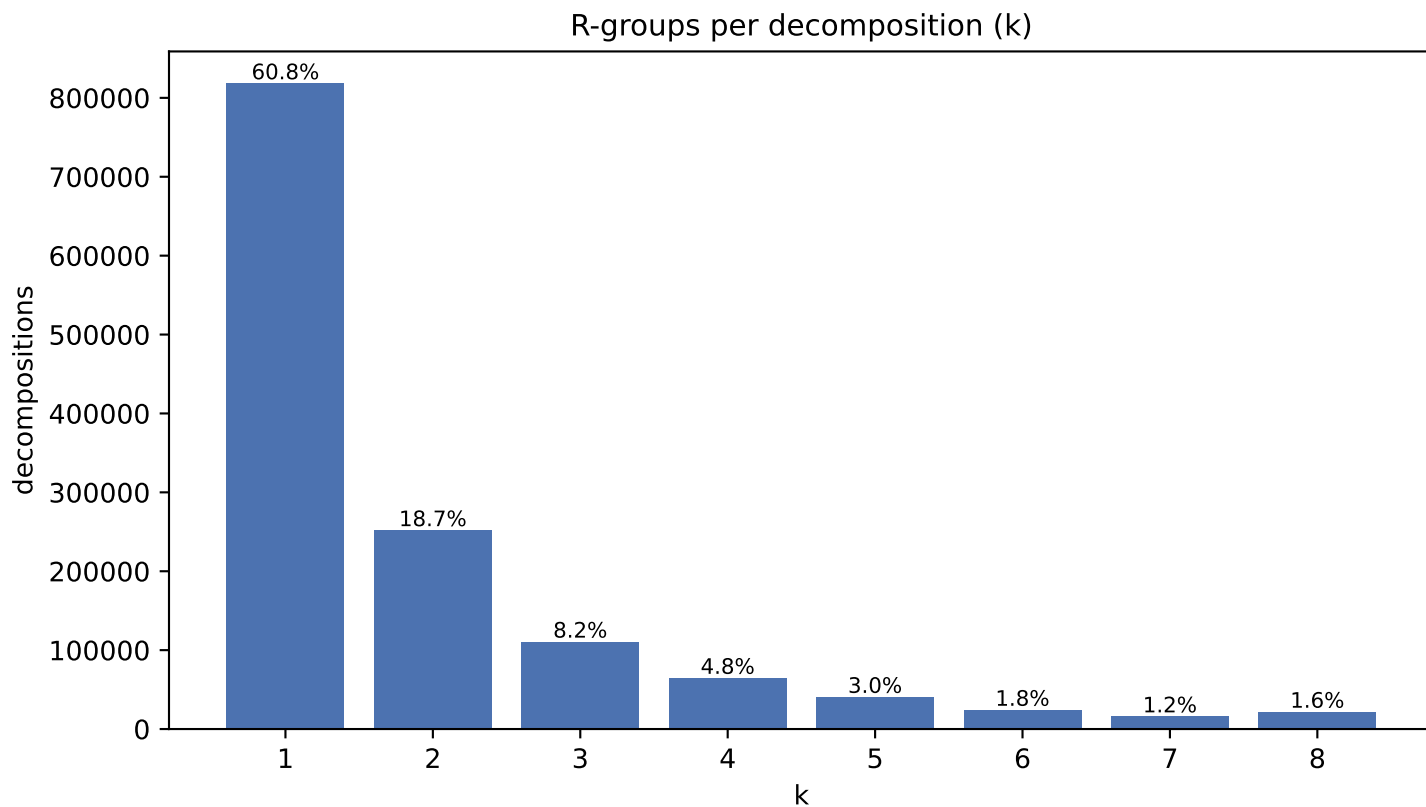


CNP0004139 k=4 core 17 atoms, R-groups [1, 1, 1, 1]

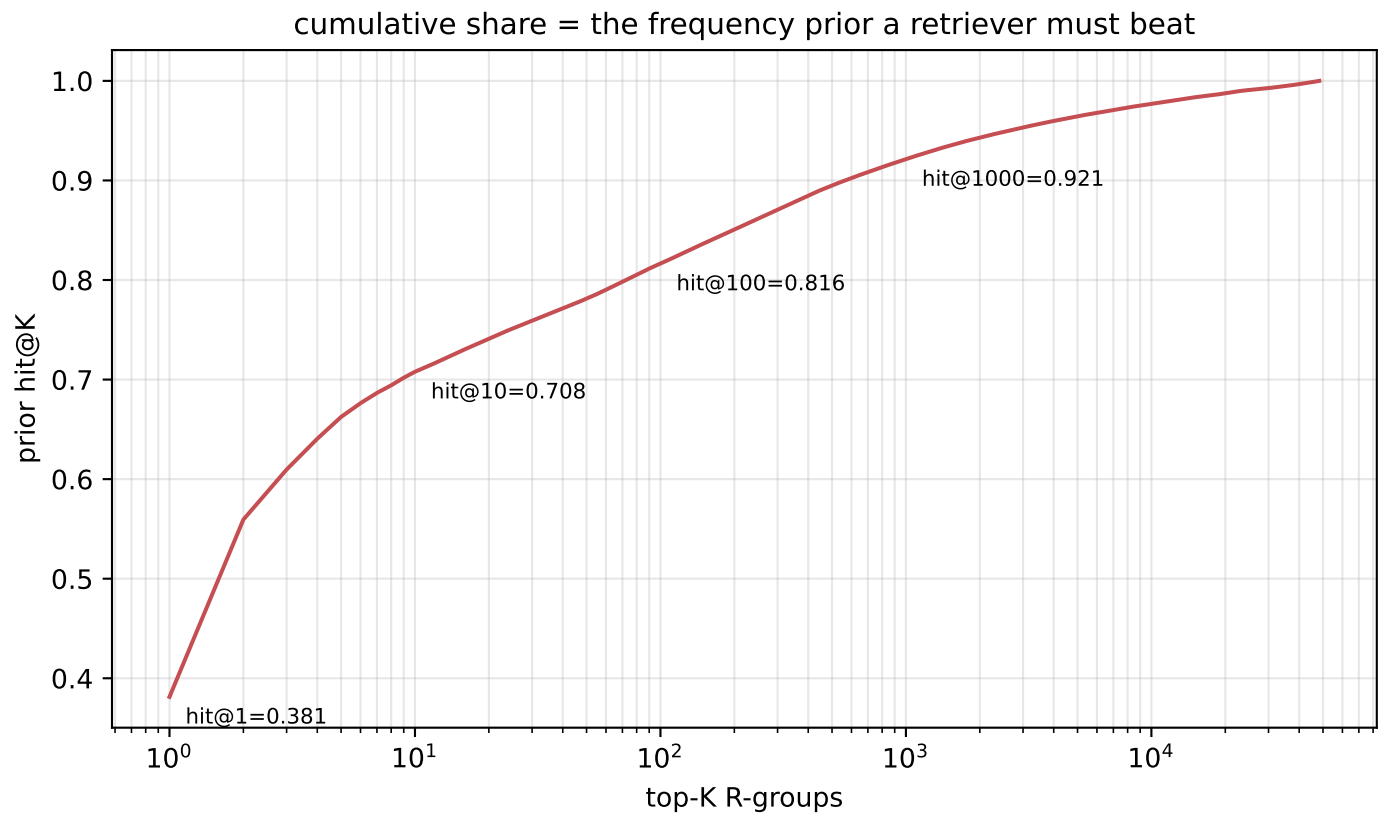
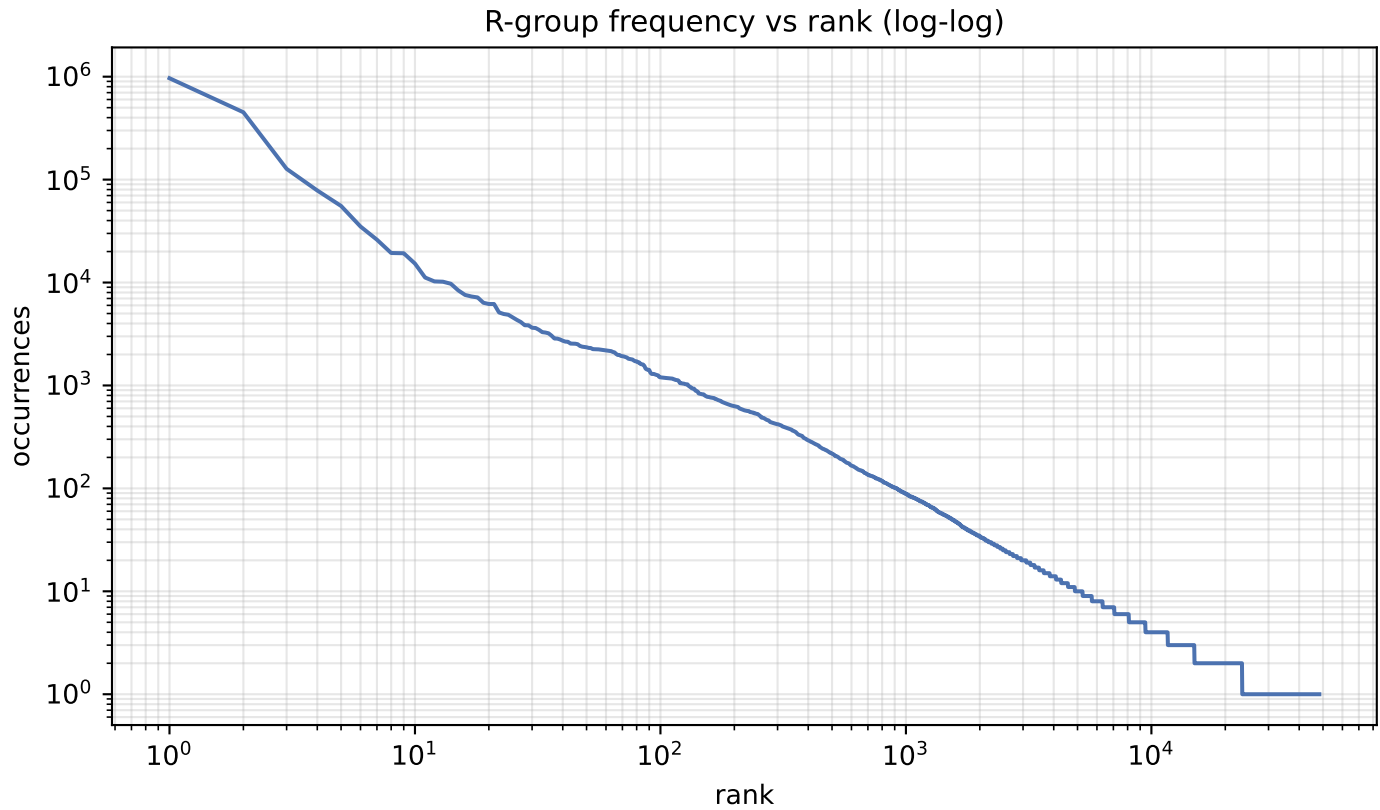


blue = core orange = R-groups red = cut bonds

F · Decomposition shape & preserved chemistry



G · R-group library skew



H · R-group library

quantity	value
distinct R-groups	48,304
occurrences	2,532,793
effective size (exp H)	46
top-1 share	38.13%
top-10 cover	70.8%
top-100 cover	81.6%

Effective size, not the raw count, is the number of classes retrieval actually chooses between.

I · Most frequent R-groups

count	share	SMILES
965,772	38.13%	*O
451,083	17.81%	*C
127,174	5.02%	*C(C)=O
78,385	3.09%	*OC
55,464	2.19%	*C(C)C
35,006	1.38%	*c1ccccc1
26,071	1.03%	*CC
19,359	0.76%	*OC(C)=O
19,250	0.76%	*Cc1ccccc1
15,272	0.60%	*N
11,179	0.44%	*C(=O)c1ccccc1
10,255	0.40%	*O[C@@H]1O[C@H](CO)[C@@H](O)[C@H](O)[C@@H]1O
10,185	0.40%	*OCC
9,732	0.38%	*c1ccc(OC)cc1
8,397	0.33%	*CO
7,577	0.30%	*CCC
7,314	0.29%	*c1ccc(O)cc1
7,163	0.28%	*Cl
6,361	0.25%	*[O-]
6,184	0.24%	*CCO

J · Instance-level common-R-group filter

quantity	value
percentile	99.9
count threshold	2,362
hashes flagged common	49 of 48,304
their share of occurrences	78.0%
single-R-group instances rejected	29,450/37,613 (78.3%)
mean R-group size, rejected	1.80 atoms
mean R-group size, kept	9.09 atoms

Filters INSTANCES, not the library: the retrieval target space and its frequency prior are unchanged. At k=1 the rule reduces to 'drop if the R-group is common', so rejection is far heavier than MolPLA saw.